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Pl o robô RPR

$$A_3^0 = \begin{bmatrix} C_1 & 0 & S_1 & 0 \\ S_1 & 0 & -C_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} C_3 & -S_3 & 0 & l_3 C_3 \\ S_3 & C_3 & 0 & l_3 S_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

 A_1^0 A_2^1 A_3^2

$$A_3^0 = \begin{bmatrix} C_{13} & -S_{13} & 0 \\ S_{13} & C_{13} & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} l_3 C_{13} + d_2 S_1 \\ l_3 S_{13} - d_2 C_1 \\ 0 \\ 1 \end{bmatrix} \quad \text{Cin Direto}$$

P_e

$$C_{13} = \cos(\theta_1 + \theta_3)$$

$$R_z(\theta_1 + \theta_3) \Rightarrow \boxed{\phi_z = \theta_1 + \theta_3}$$

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$$\begin{cases} P_x = l_3 C_{13} + d_2 S_1 = f_x(\theta_1, d_2, \theta_3) \\ P_y = l_3 S_{13} - d_2 C_1 = f_y(\theta_1, d_2, \theta_3) \\ \phi_2 = \theta_1 + \theta_3 = f_{\phi_2}(\theta_1, d_2, \theta_3) \end{cases}$$

$$\frac{\partial f_x}{\partial \theta_1} = -l_3 S_{13} + d_2 C_1$$

$$\frac{\partial f_x}{\partial d_2} = S_1$$

$$\frac{\partial f_x}{\partial \theta_3} = -l_3 S_{13}$$

$$\frac{\partial f_y}{\partial \theta_1} = l_3 C_{13} + d_2 S_1$$

$$\frac{\partial f_y}{\partial d_2} = -C_1$$

$$\frac{\partial f_y}{\partial \theta_3} = l_3 C_{13}$$

$$\frac{\partial \phi_2}{\partial \theta_1} = 1$$

$$\frac{\partial \phi_2}{\partial d_2} = 0$$

$$\frac{\partial \phi_2}{\partial \theta_3} = 1$$

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 J_{P_1} J_{P_2} J_{P_3}

3

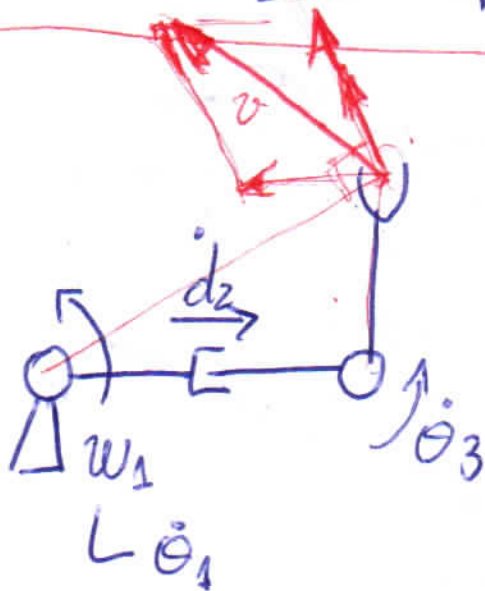
$$\begin{bmatrix} \dot{P}_x \\ \dot{P}_y \\ \dot{w}_z \end{bmatrix} = \begin{bmatrix} -l_3 s_{13} + d_2 c_1 & s_1 & -l_3 s_{13} \\ l_3 c_{13} + d_2 s_1 & -c_1 & l_3 c_{13} \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{d}_2 \\ \dot{\theta}_3 \end{bmatrix}$$

$J \Rightarrow \text{Jacobiano}$

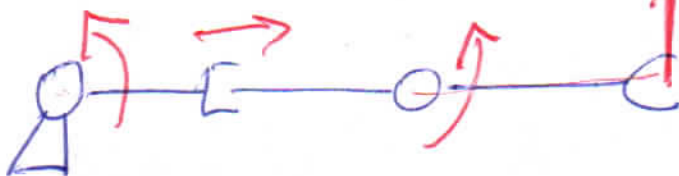
$$\vec{v} = J \dot{q}$$

$\hookrightarrow J(q)$

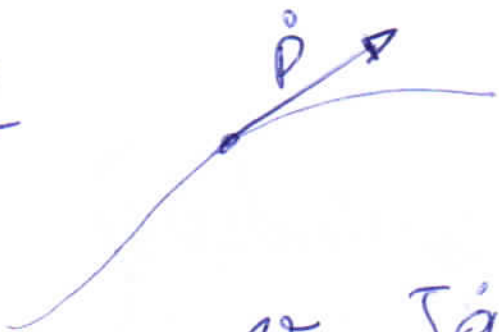
Ex



$$\begin{aligned} \dot{\theta}_1 &= 1 \text{ rad/s} \\ \Rightarrow \dot{d}_2 &= 0 \text{ mm/s} \\ \dot{\theta}_3 &= 1 \text{ rad/s} \end{aligned}$$



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(4)

$$v = J\dot{q}, \text{ então}$$

$$\boxed{\dot{q} = J^{-1}v} \quad \text{Cin. Dif. inversa}$$

↑
inversa do Jacobiano

$\Rightarrow J \Rightarrow$ Posto completo

$\hookrightarrow$ é invertível

$\hookrightarrow$ LI

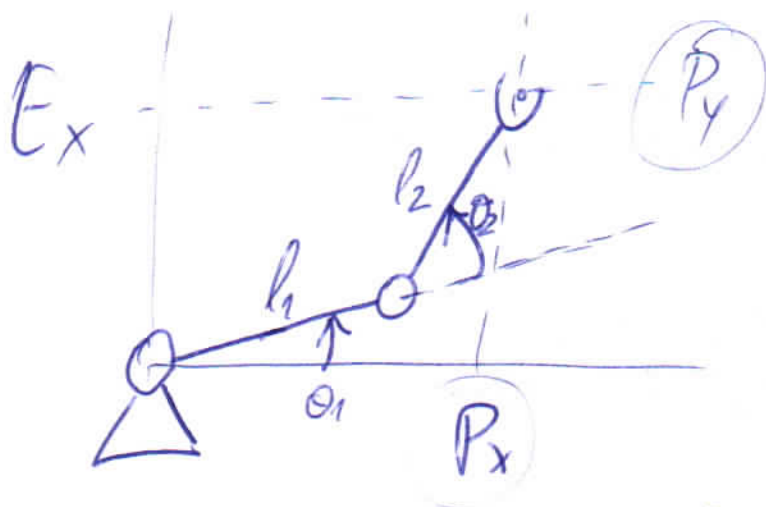
Se $J(q)$ perde Posto

$\hookrightarrow$ LD

Se $J(q) \Rightarrow$ LD então: $\det(J) = 0$

$J(q) \Rightarrow$ LI então: $\det(J) \neq 0$

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$$\begin{cases} P_x = l_1 c_1 + l_2 c_{12} \\ P_y = l_1 s_1 + l_2 s_{12} \end{cases}$$

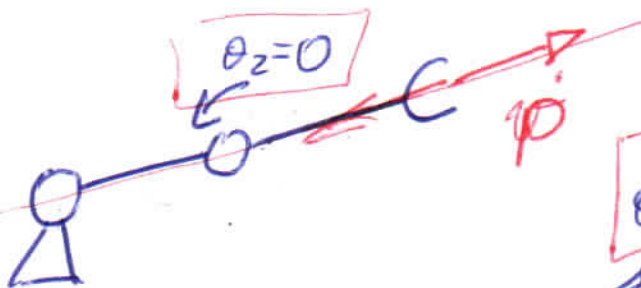
$$J = \begin{bmatrix} -l_1 s_1 - l_2 s_{12} & -l_2 s_{12} \\ l_1 c_1 + l_2 c_{12} & l_2 c_{12} \end{bmatrix} \checkmark$$

$$\begin{bmatrix} \dot{P}_x \\ \dot{P}_y \end{bmatrix} = J \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

$$\dot{q} = J^{-1} v$$

$$\det(J) = S_2 = \sin(\theta_2) = 0$$

$$\sin(\theta_2) = 0 \Rightarrow \theta_2 = 0 \text{ rad ou } \pi \text{ rad}$$



$$\theta_2 = \pi$$

Posturas

Singulares

$\det(J) = 0$
Cálculo de
Singularidades

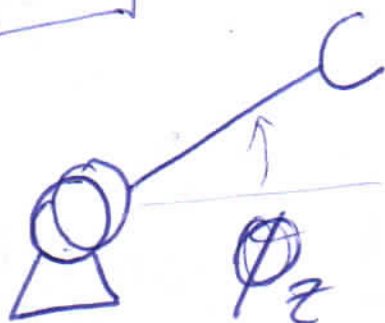
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P/ RPR

$$\boxed{\det(J) = d_z}$$

$$\boxed{d_z = 0}$$



$$\boxed{\phi_z = \theta_1 + \theta_3}$$

infinitas soluciones

Jacobiano Geométrico

Caso Espacial

$$\begin{cases} P_x = f_x \\ P_y = f_y \\ P_z = f_z \end{cases}$$

$$\begin{cases} \phi_x \\ \phi_y \\ \phi_z \end{cases} \begin{cases} f_{\phi_x} \\ f_{\phi_y} \\ f_{\phi_z} \end{cases} \begin{matrix} X & X \\ \text{Problema.} \end{matrix}$$

3/3 Jacobiano Geométrico (7)

$$v = J \dot{q}$$

$$\begin{bmatrix} \dot{p}_x \\ \dot{p}_y \\ \dot{p}_z \\ \vdots \\ w_x \\ w_y \\ w_z \end{bmatrix} = \begin{bmatrix} \dot{p} \\ \vdots \\ w \end{bmatrix} = \begin{bmatrix} J_{p1} & J_{p2} & \dots & J_{pn} \\ \vdots & \vdots & \ddots & \vdots \\ J_{o1} & J_{o2} & \dots & J_{on} \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \vdots \\ \dot{q}_n \end{bmatrix}$$

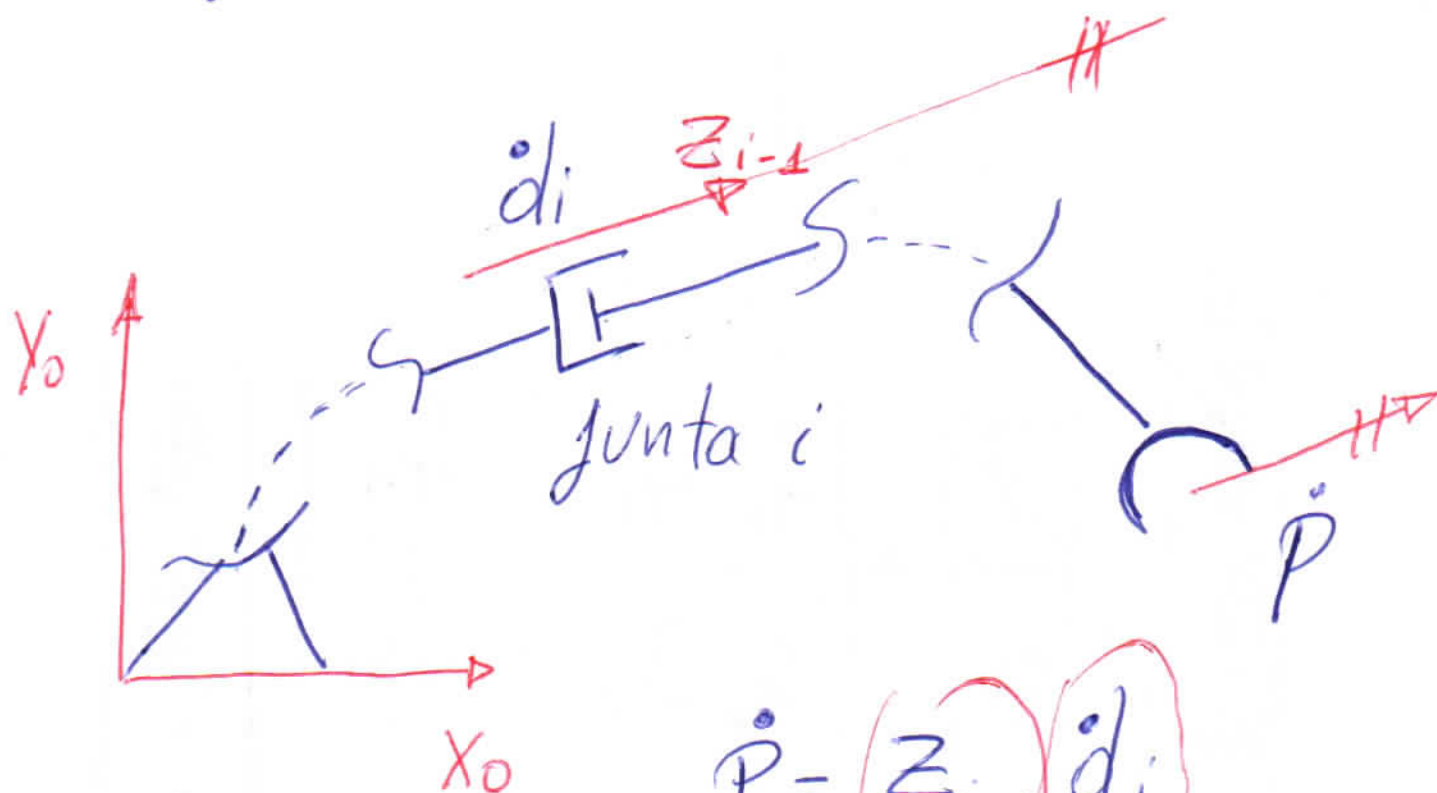
$J_{p_i} \Rightarrow$ Contribuição da junta i na vel. linear do efet.

$J_{o_i} \Rightarrow$ Contribuição da junta i na vel. angular do efet.

J_{p_i} e J_{o_i} ?

3/2 P/ juntas prismáticas

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Contrib
Velocidade
linear.

$$\dot{P} = Z_{i-1} \dot{d}_i$$

$$\dot{P} = J_{P_i} \dot{q}_i$$

$$J_{P_i} = Z_{i-1}$$

$Z_{i-1} \Rightarrow$ eixo Z do sist. coord O_{i-1}
em relação à base

$$Z_{i-1}^0 \Rightarrow Z_{i-1}$$

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$$R_{i-1}^0 = R_1^0 R_2^1 \dots R_{i-2}^{i-3} R_{i-1}^{i-2}$$

$$= \begin{bmatrix} | & | & | \\ X_{i-1} & Y_{i-1} & Z_{i-1} \\ | & | & | \end{bmatrix}$$

$$A_{i-1}^0 = \begin{bmatrix} R & P \\ \infty & 1 \end{bmatrix}$$

Contribuições na velocidade angular

$$w = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = J_{0,i} \dot{q}_i = J_{0,i} \dot{d}_i$$

↑
não contribui

→ $\neq 0$

→ $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$J_{0,i} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}_{3 \times 1}$$

$$J_{P,i} = \begin{bmatrix} 1 \\ Z_{i-1} \\ 1 \end{bmatrix}_{3 \times 1}$$